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June 6, 2026 – Paestum, Salerno

Introduction to OpenFOAM® Computational Library and Design of Profile Extrusion Dies

P1 - Introduction to OpenFOAM and Simulation Fundamentals

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Outline

10:00 – 11:15	Introduction to OpenFOAM and Simulation Fundamentals (P1)
11:30 – 13:00	Geometry, Mesh Generation, and Case Setup in OpenFOAM (P2)
13:00 – 15:00	Lunch break
15:00 – 17:00	Hands-On Profile Extrusion Dies (P3)



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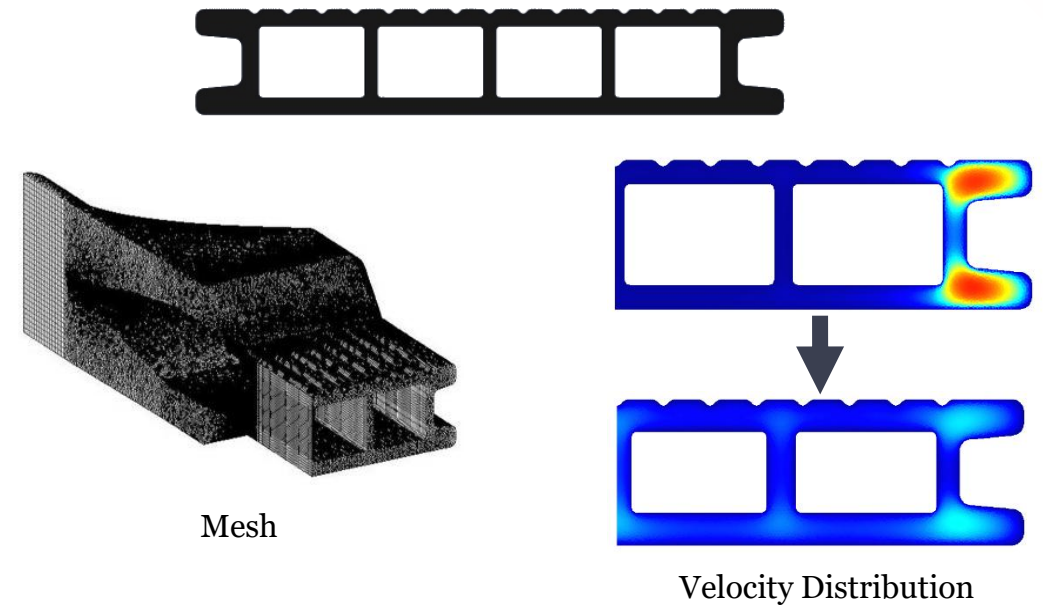
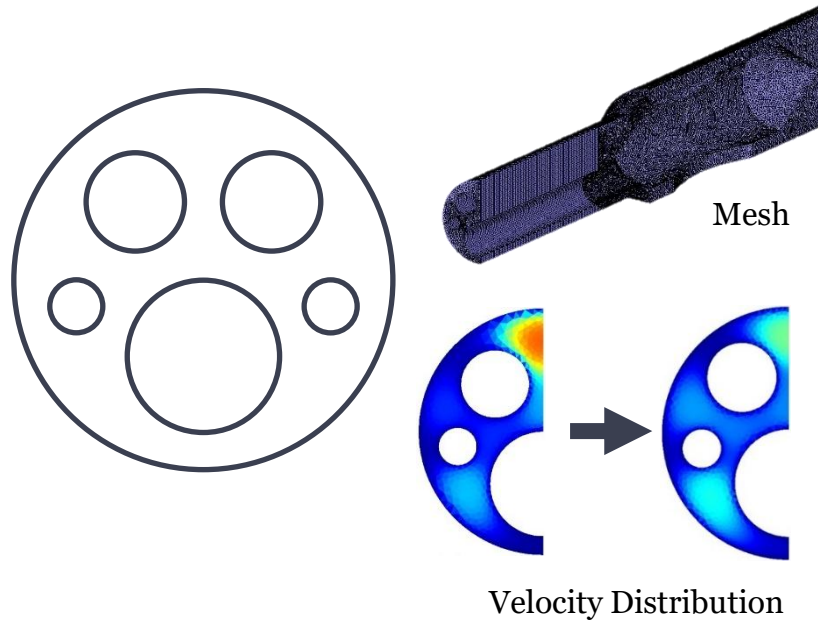


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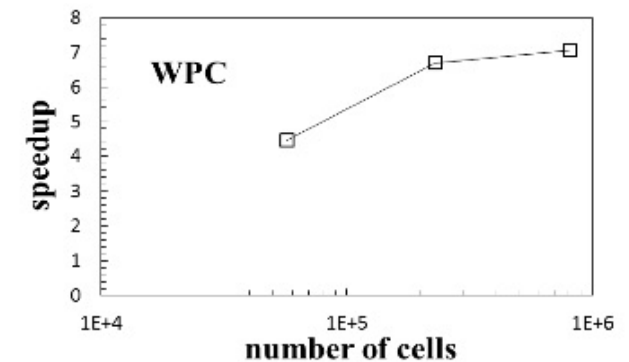
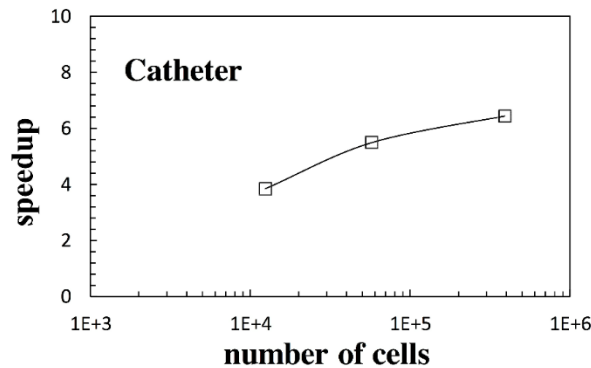
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Motivation

Unstructured Numerical Modeling Code



GPU Parallelization



Motivation

New feature → New researcher



Hussaini Hanging Bridge, Pakistan



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The OpenFOAM® Computational Library

A (very) brief history...

1996 – Hrvoje Jasak PhD Thesis, Imperial College

1998 – Weller, Tabor, Jasak, Fureby, Computers in Physics

A tensorial approach to computational continuum mechanics using object-oriented techniques

H. G. Weller and G. Tabor^{ab}
Department of Mechanical Engineering, Imperial College, London SW7 2BX, United Kingdom
H. Jasak
Computational Dynamics Limited, London W10 6RA, United Kingdom
C. Fureby
Department of Weapons and Protection, National Defense Research Establishment (FOA), S-17290 Stockholm, Sweden

2006 – 1st OpenFOAM Workshop (Zagreb)

2010 – Foam-Extend v1.6

2013 – 1st OpenFOAM Conference (Frankfurt)

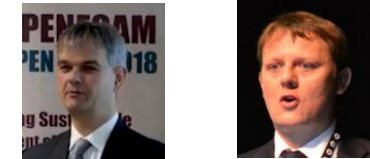
2016 – 11th OpenFOAM Workshop (Guimarães)
4th OpenFOAM Conference (Cologne)

2016 – Discussions about “One OpenFOAM”
Creation of OpenFOAM Wiki



2021 – The OpenFOAM Journal
OpenFOAM®
Journal

1989 – FOAM v1 – Imperial College
Henry Weller and Hrvoje Jasak



2000 – Nabla Ltd. Foundation
FOAM released as commercial CFD Toolbox

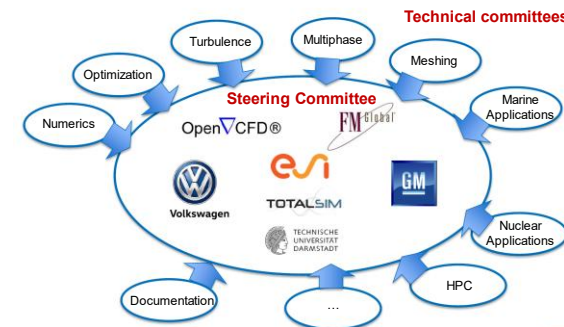
2004 – Nabla Ltd. Closure
OpenFOAM v1.0 released as open-source by OpenCFD

2006 – Trademarks OpenFOAM and OpenCFD registration (to avoid misuse)

2011 – Acquisition of OpenCFD by SGI and creation of OpenFOAM Foundation (to assure the open-source nature)

2012 – Acquisition of OpenCFD by ESI

2018 – OpenFOAM Governance



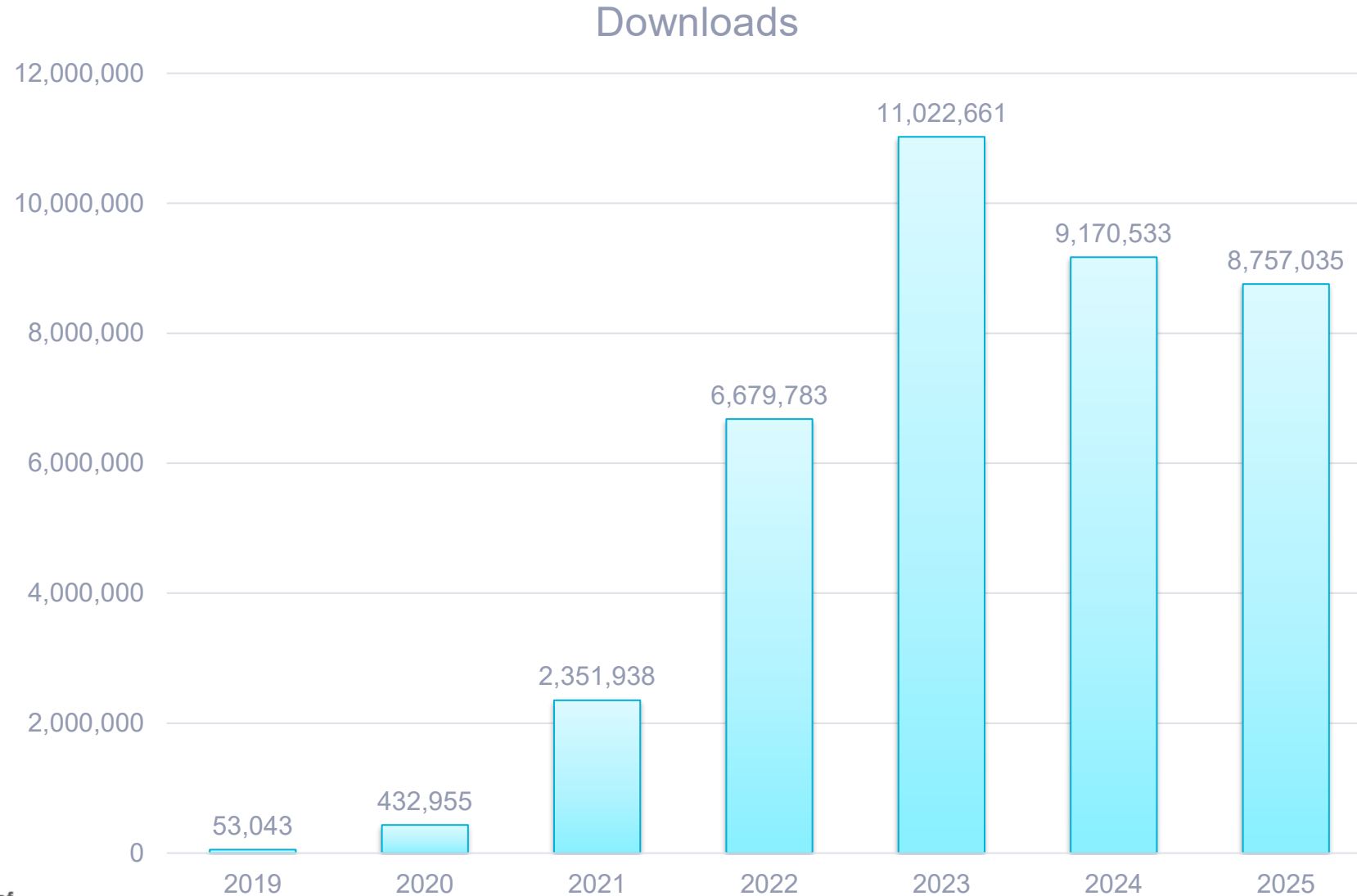
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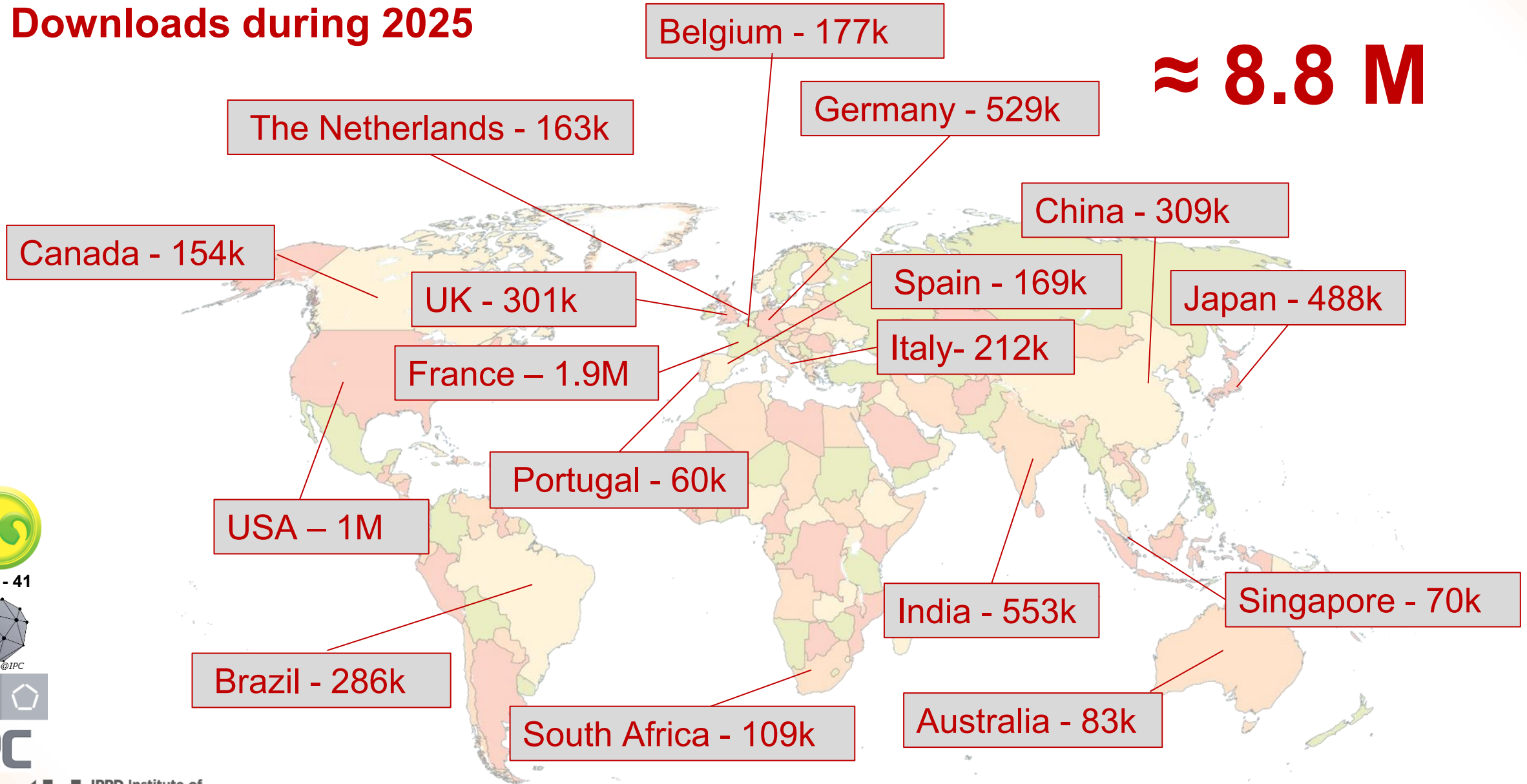
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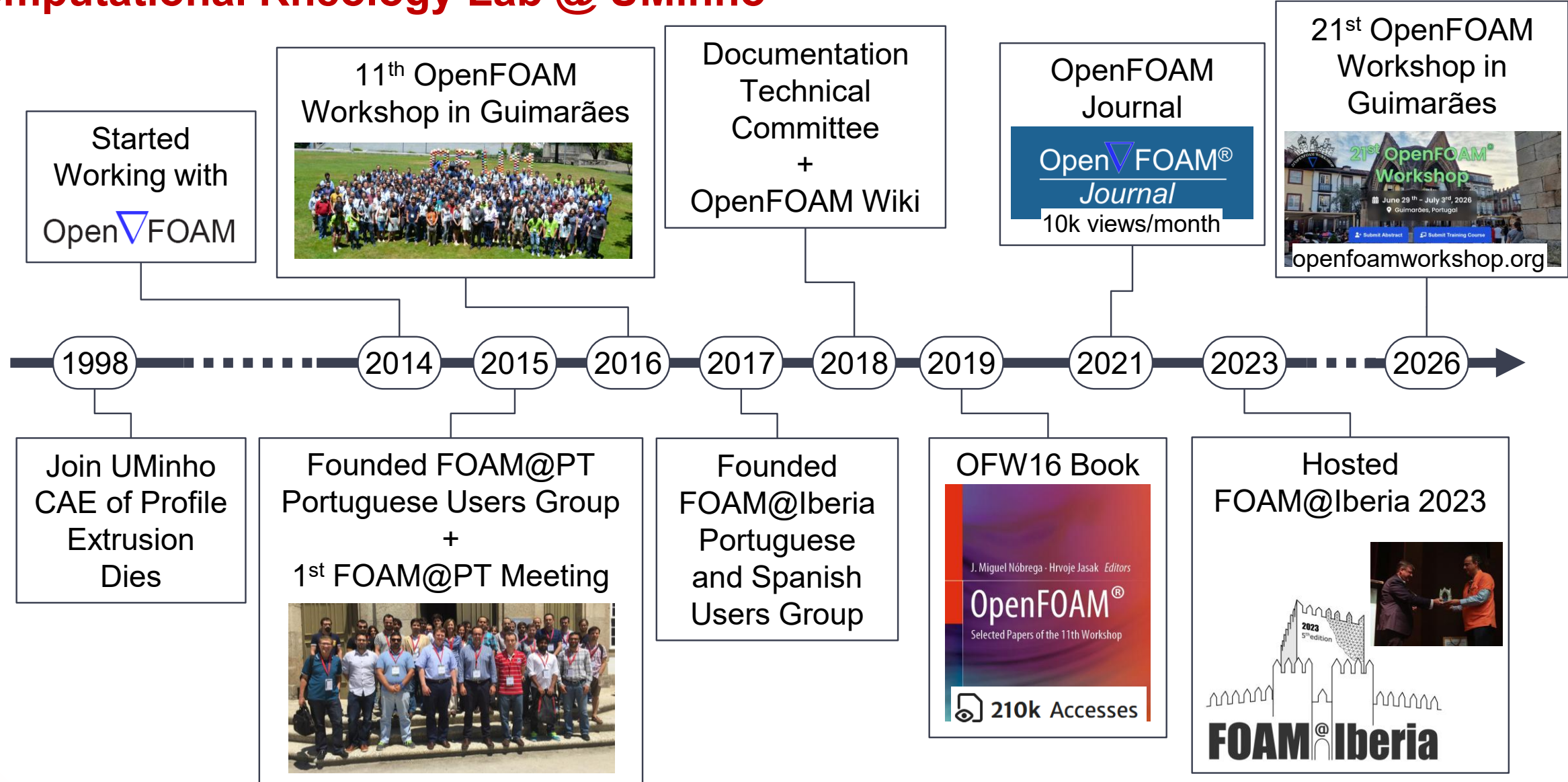
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Computational Rheology Lab @ UMinho



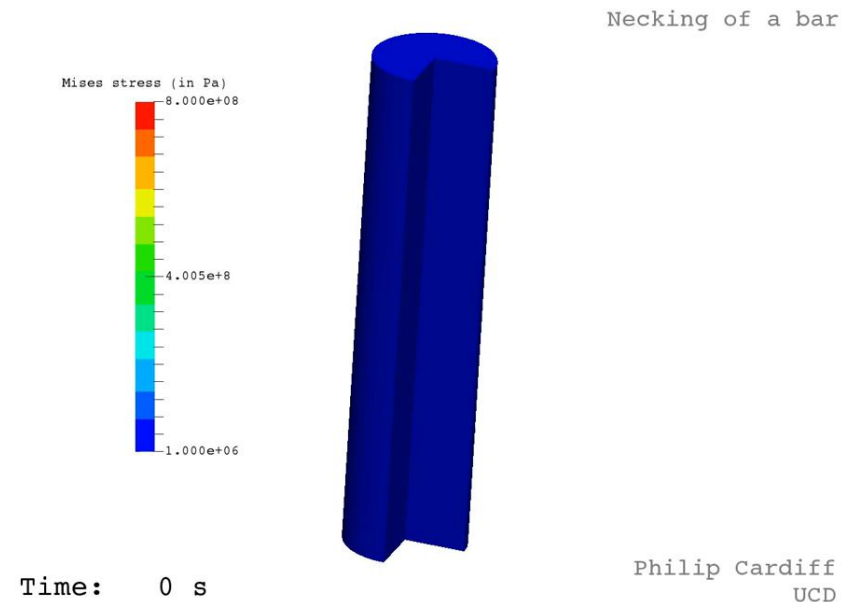
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Open▽FOAM

Fluid Dynamics



Solid Mechanics



Necking of a bar

Fluidized Bed (Eulerian+Lagrangian)

From Philip Cardiff's Youtube channel - <https://tinyurl.com/pcardiff>

The OpenFOAM® Computational Library

Equation mimicking

$$\rho \frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) - \mu \nabla^2 \mathbf{U} = -\nabla p$$

solve

```
(  
    fvm::ddt(rho,U) + fvm::div(phi,U) - fvm::laplacian(mu,U) == -fvc::grad(p)  
);
```



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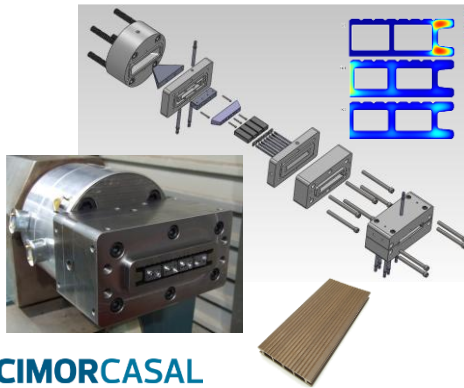
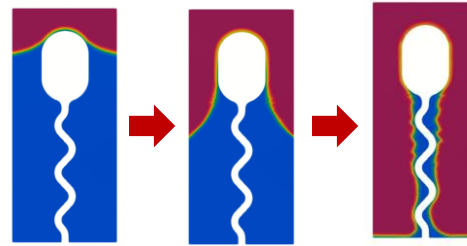
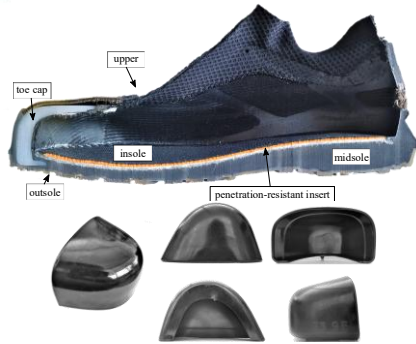
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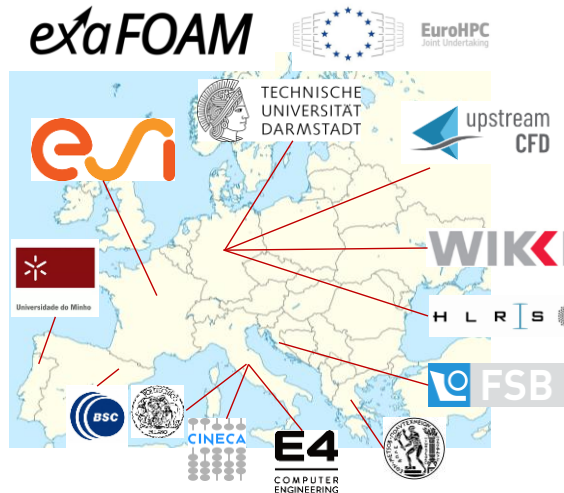
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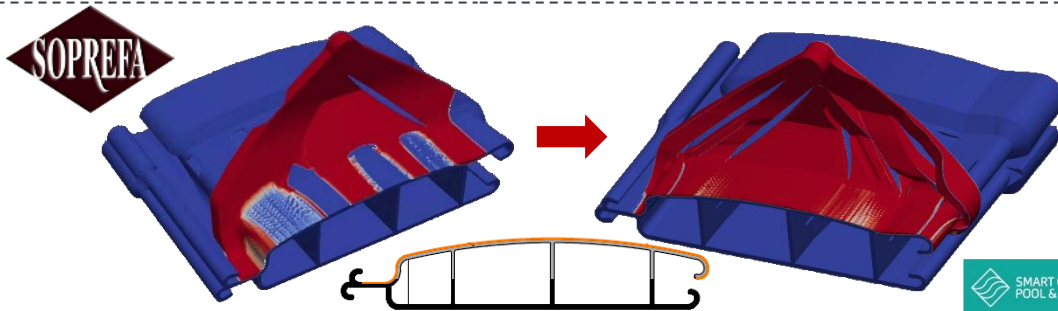
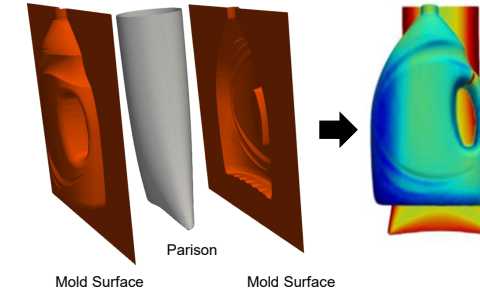
FAMEST
— Footwear Advanced Materials, Experiments
and Software Technologies —



SOCIMORCASAL



LOGOPLASTE

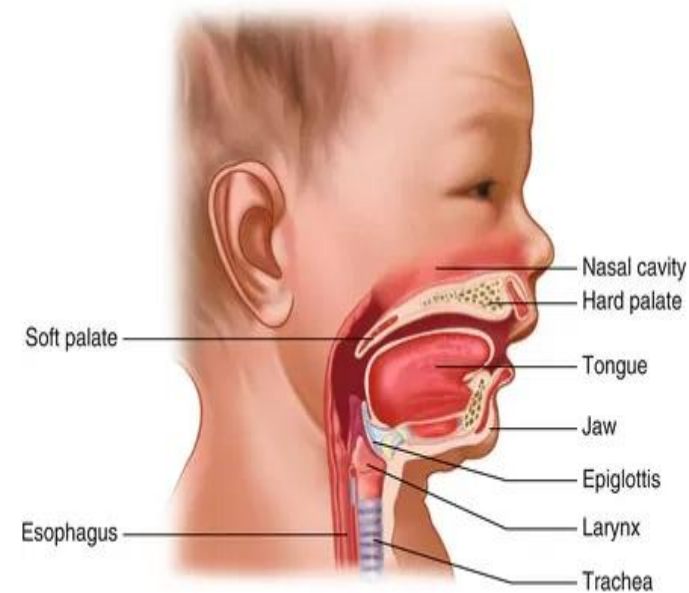




Introduction

Pacifiers Assessment

Development of a computational methodology capable of predicting the newborn oral cavity behavior during pacifier sucking



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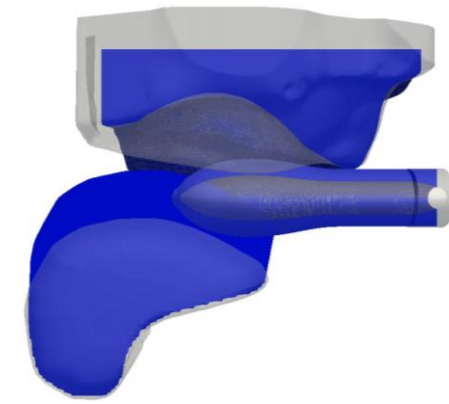
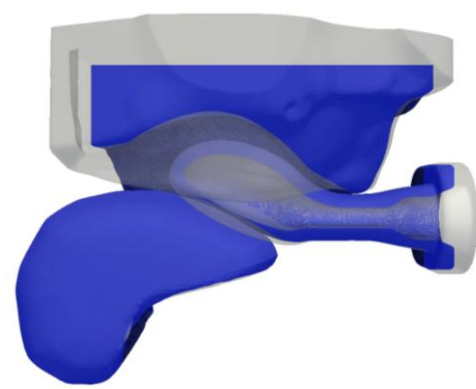
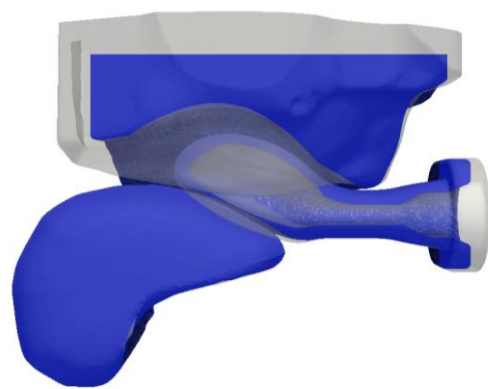
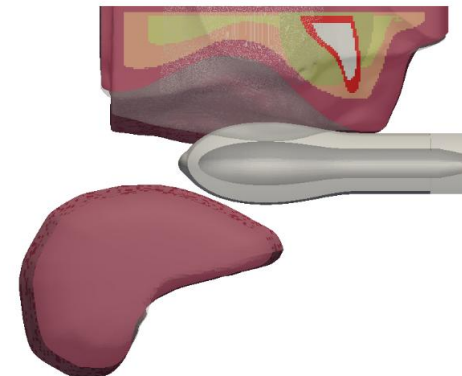
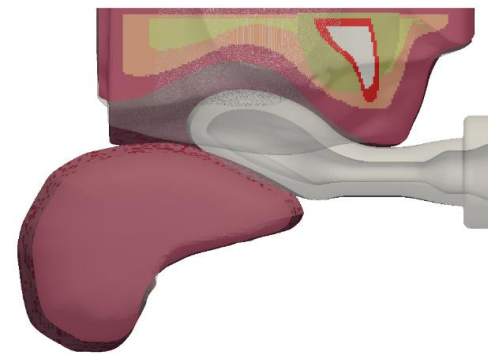
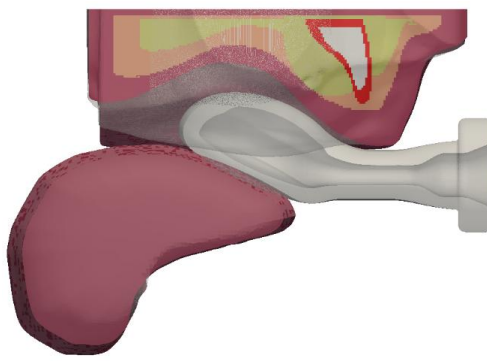


Pacifiers Assessment

NUK Genius Pacifier

NUK Standard Pacifier

Conventional Pacifier



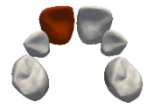
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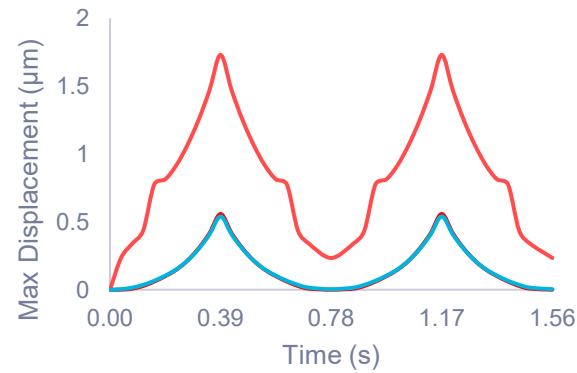
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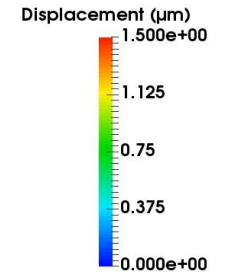
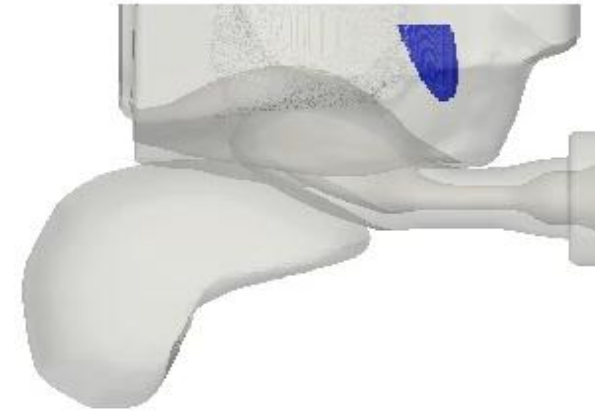
Pacifiers Assessment



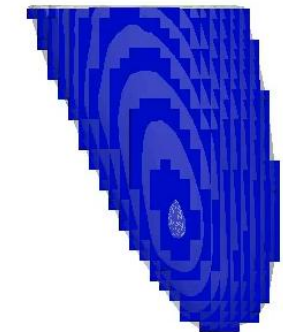
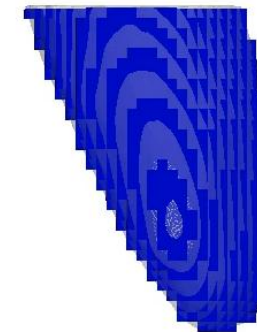
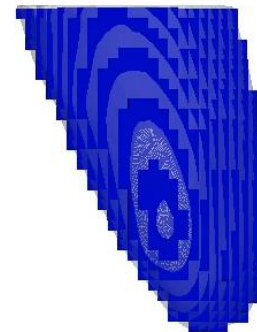
Central Incisor



- NUK Genius Pacifier
- NUK Standard Pacifier
- Conventional Pacifier



Conventional Pacifier



Displacement 1000x

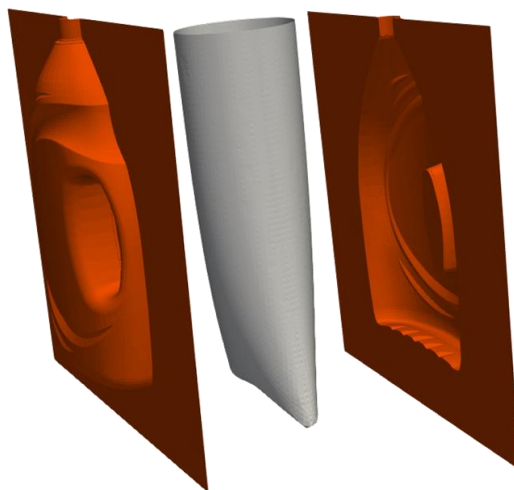


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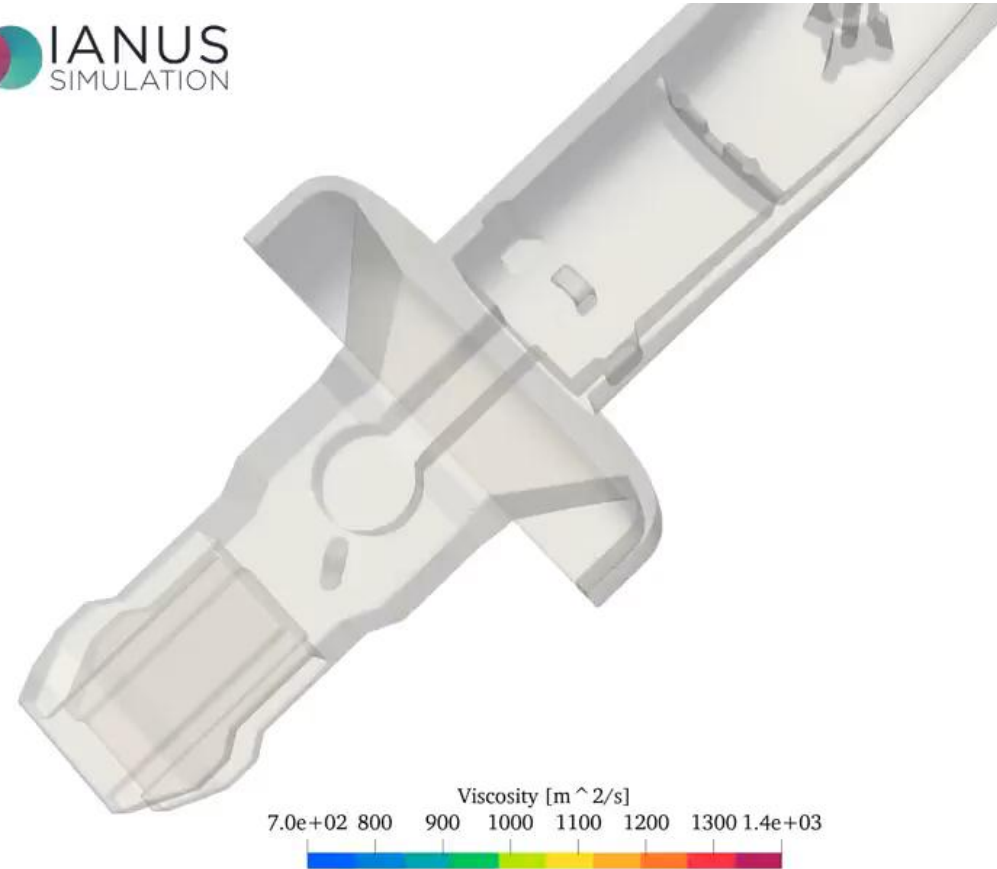
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Extrusion Blow Molding Simulator



Introduction

Injection Molding



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OpenFOAM-preCICE: Coupling OpenFOAM with External Solvers for Multi-Physics Simulations

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DOI: <https://doi.org/10.51560/ofj.v3.88>

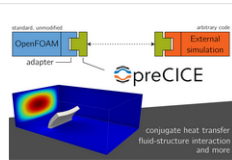
Keywords: Fluid Structure Interaction, conjugate heat transfer, multiphysics simulation

Abstract

Multi-physics simulations, such as conjugate heat transfer or fluid-structure interaction, are often constructed completely in OpenFOAM. However, they can also be formed by coupling OpenFOAM to third-party simulation software via a coupling tool. This approach indirectly adds to the capabilities of OpenFOAM those of other simulation tools (such as physical models or discretization methods more fitting for specific applications), and allows building complex multi-physics simulations by connecting specialized single-physics codes. We present the OpenFOAM-preCICE adapter, a function object that enables standard OpenFOAM solvers to use the open-source, massively parallel coupling library preCICE, without requiring any code modifications. We review alternative coupling approaches, analyze our design decisions, peek into key implementation details, validate the adapter, study the effect on runtime, and give an overview of the growing community of users and contributors.

Author Biography

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PDF Discussion Forum
Abstract Video

Case files and source code

Published
2023-02-27

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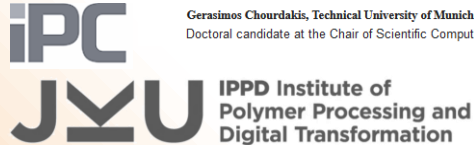
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ancolli
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Alejandro
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I have read your article and it is really interesting. I have two minor questions:

a) In the partitioned and conjugated heat transfer examples, you have used a Dirichlet-Neuman coupling. A few years ago, for a similar problem, I found that this was a faster method than the harmonic averaging already implemented in the chtMultiregionFoam. However, with the Dirichlet-Neuman coupling, I found some convergence problems depending on the relative difference of the conductivities on each side (electrical problems).



OpenFOAM-preCICE: Coupling OpenFOAM with External Solvers for Multi-Physics Simulations

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Author Biography

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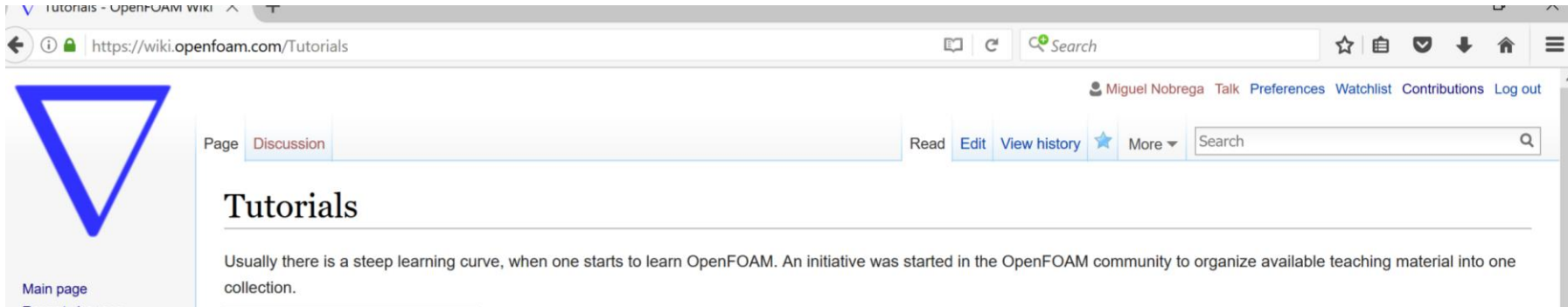
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Discussion Forum

More Fun with OpenFOAM?

OpenFOAM Wiki - wiki.openfoam.com



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Computational Modelling

Notation

- Scalars – plain characters - e.g. p, ρ, η_p, λ
- Tensors [], vectors { } or bold characters – e.g. $\mathbf{U}, \boldsymbol{\tau}$
- Nabla (∇) the following differential operator

$$\nabla = \begin{pmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{pmatrix}$$



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Computational Modelling

Governing Equations

- Continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0$$

- Momentum

$$\frac{\partial (\rho \mathbf{U})}{\partial t} + \nabla \cdot (\rho \mathbf{U} \mathbf{U}) + \nabla \cdot (p \mathbf{I}) + \nabla \cdot \boldsymbol{\tau} = 0$$

- Constitutive Model (Oldroyd-B)

$$\boldsymbol{\tau} = \boldsymbol{\tau}_S + \boldsymbol{\tau}_P$$

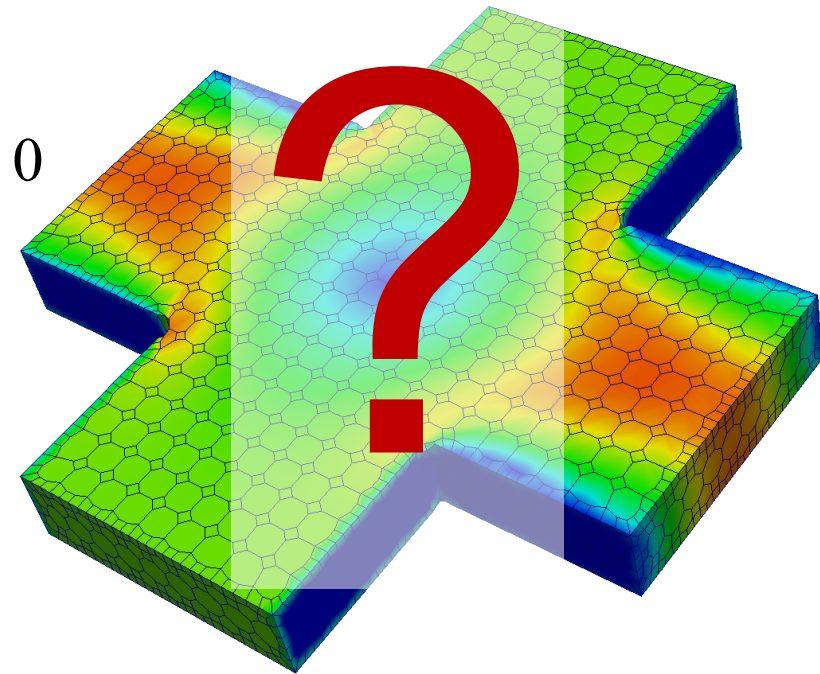
Solvent

$$\boldsymbol{\tau}_S = \eta_S \left(\nabla \mathbf{U} + (\nabla \mathbf{U})^T \right)$$

Polymeric

$$\boldsymbol{\tau}_P + \lambda \left(\frac{\partial \boldsymbol{\tau}_P}{\partial t} + \nabla \cdot (\mathbf{U} \boldsymbol{\tau}_P) \right) = \eta_P \left(\nabla \mathbf{U} + \nabla \mathbf{U}^T \right) + \lambda \left(\boldsymbol{\tau}_P \cdot \nabla \mathbf{U} + \boldsymbol{\tau}_P \cdot (\nabla \mathbf{U})^T \right)$$

Velocity magnitude distribution



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Computational Modelling

- Continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{U}) = 0$$

- Momentum

$$\frac{\partial (\rho \phi)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \phi) - \nabla \cdot (\rho \Gamma_{\phi} \nabla \phi) = S_{\phi}(\phi)$$

Governing Equations

Velocity magnitude distribution

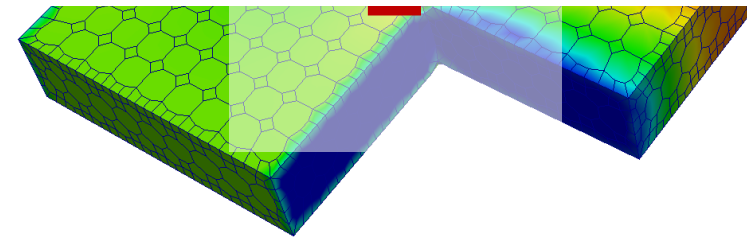


Solvent

$$\boldsymbol{\tau}_S = \eta_S (\nabla \mathbf{U} + (\nabla \mathbf{U})^T)$$

Polymeric

$$\boldsymbol{\tau}_P + \lambda \left(\frac{\partial \boldsymbol{\tau}_P}{\partial t} + \nabla \cdot (\mathbf{U} \boldsymbol{\tau}_P) \right) = \eta_P (\nabla \mathbf{U} + \nabla \mathbf{U}^T) + \lambda \left(\boldsymbol{\tau}_P \cdot \nabla \mathbf{U} + \boldsymbol{\tau}_P \cdot (\nabla \mathbf{U})^T \right)$$



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Computational Modelling

The Standard Governing Equation

$$\underbrace{\frac{\partial(\rho\phi)}{\partial t}}_{\text{Time evolution}} + \underbrace{\nabla \cdot (\rho \mathbf{U} \phi)}_{\text{Advection}} - \underbrace{\nabla \cdot (\rho \Gamma_{\phi} \nabla \phi)}_{\text{Diffusion}} = \underbrace{S_{\phi}(\phi)}_{\text{Source}}$$



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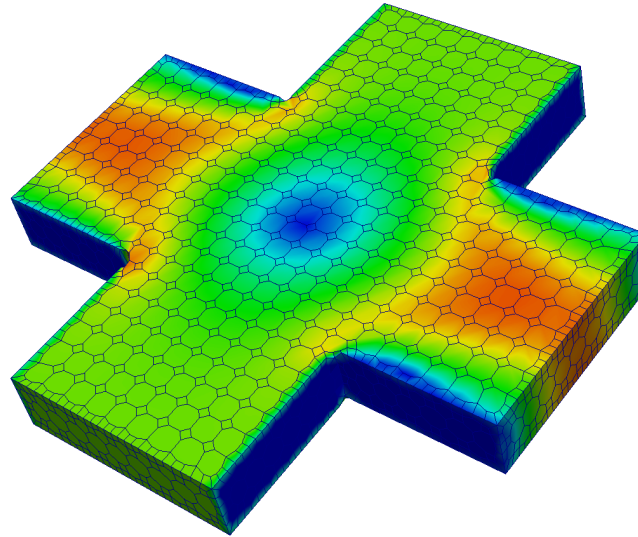
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The Standard Governing Equation

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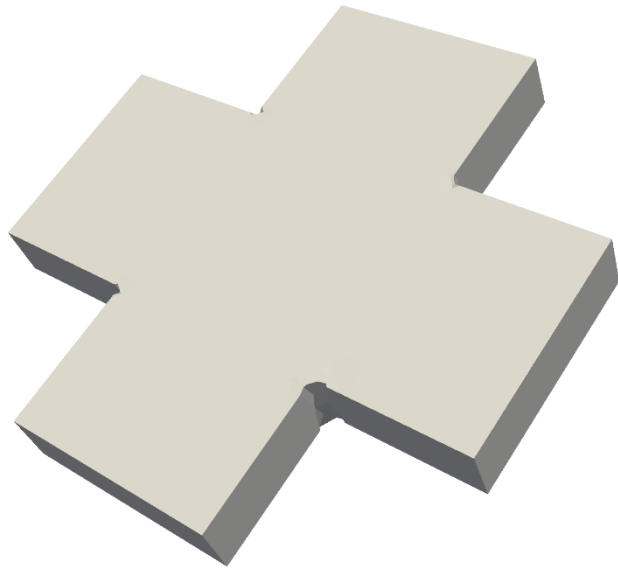
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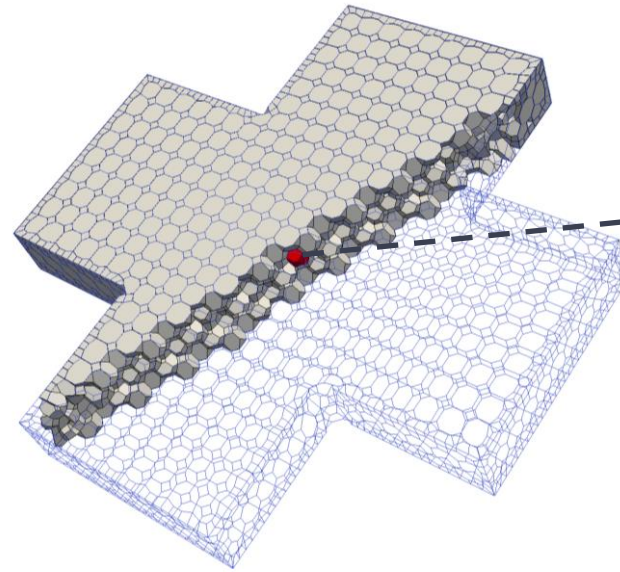
Computational Modelling

The Finite Volume Method

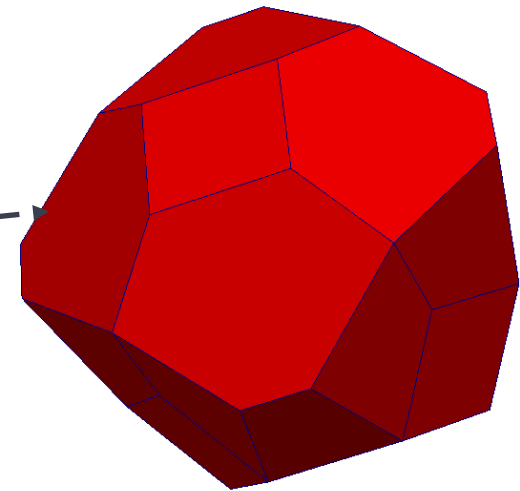
$$\frac{\partial(\rho\phi)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \phi) - \nabla \cdot (\rho \Gamma_{\phi} \nabla \phi) = S_{\phi}(\phi)$$



Geometry



Mesh



Cell



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Computational Modelling

The Finite Volume Method

$$\frac{\partial(\rho\phi)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \phi) - \nabla \cdot (\rho \Gamma_{\phi} \nabla \phi) = S_{\phi}(\phi)$$



$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & a_{21} & \dots & a_{21} \\ \dots & \dots & \dots & \dots \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{bmatrix} \begin{Bmatrix} \phi_1 \\ \phi_2 \\ \dots \\ \phi_N \end{Bmatrix} = \begin{Bmatrix} b_1 \\ b_2 \\ \dots \\ b_N \end{Bmatrix}$$



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Computational Modelling

The Finite Volume Method

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$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1N} \\ a_{21} & & & \\ \dots & & & \\ a_{N1} & a_{N2} & \dots & a_{NN} \end{bmatrix} \begin{bmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_N \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_N \end{bmatrix}$$

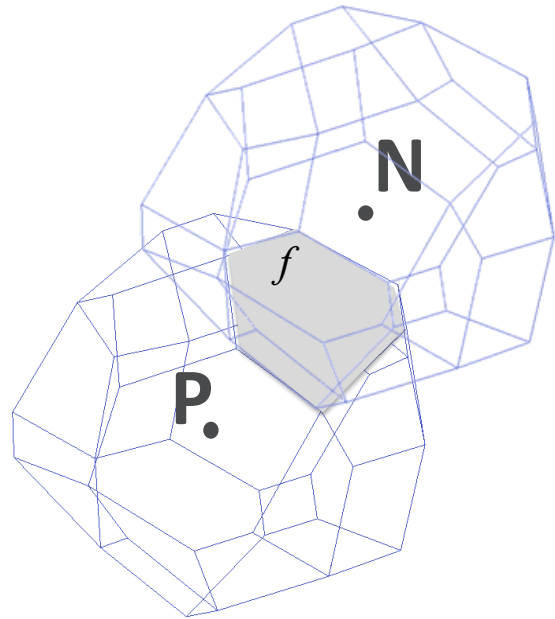
$[A] \{\phi\} = \{b\}$



Computational Modelling

The Finite Volume Method

$$\frac{\partial(\rho\phi)}{\partial t} + \nabla \cdot (\rho \mathbf{U} \phi) - \nabla \cdot (\rho \Gamma_{\phi} \nabla \phi) = S_{\phi}(\phi)$$



Cell P and N share face f

Discretization (fvSchemes)

$$\frac{\phi_N - \phi_P}{|d|} = \frac{-1}{|d|} \phi_P + \frac{1}{|d|} \phi_N$$

Add to P cell equation



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Computational Modelling

System of Equations

Solve the system(s) of equations (**fvSolution**)

$$[A]\{\phi\} = \{b\}$$

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System of Equations

Solve the system(s) of equations (fvSolution)

$$[A]\{\phi\} = \{b\}$$

- Direct Solvers
- Iterative Solvers

Computational Modelling

System of Equations – Iterative Solvers

$$[A]\{\phi\} - \{b\} = \{R\}$$

- $|R|$ quantifies the error of the current solution
 - Relative tolerance
 - Absolute Tolerance

Linux/OpenFOAM - Memory Aid

General

- ☐  WSL App ## Open Ubuntu Terminal
- ☐  ParaView App ## open Paraview for post processing
- ☐ right mouse button ## paste clipboard contents in command line
- ☐ tab key ## complete commands
- ☐ Ctrl+R ## Repeat previous Commands
- ☐ Arrow up or down ## Browse by previous commands
- ☐ >> code . ## open VSCode in the current folder
- ☐ >> shopt -s direxand #allow tab in WSL
- ☐ >> explorer . ## Open the current folder in windows explorer

Linux

- ☐ >> cd <name> ## Change to folder <name>
- ☐ >> cd .. ## Change to previous folder
- ☐ >> cd ## Change to home folder
- ☐ >> pwd ## print current (working) directory
- ☐ >> rm <file> ## remove file named <file>
- ☐ >> rm -rf <folder> ## remove folder named <folder>
- ☐ >> chmod +x <file> ## make the <file> executable
- ☐ >> touch x.foam ## create empty file named x.foam

OpenFOAM

- ☐ >> openfoam2512 ## load OpenFOAM variables
- ☐ \$FOAM_TUTORIALS ## Folder for tutorial cases
- ☐ \$FOAM_RUN ## folder to run cases
- ☐ >> tut ## change to tutorial folder
- ☐ >> run ## change to run folder
- ☐ >> <solverName> ## run solver named <solverName>
- ☐ >> <solverName> > log & ## run solver named <solverName> in background and send output to log file



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OpenFOAM 1st case study

1. Open WSL
2. Load OpenFOAM variables
3. `>> run`
4. `>> wget https://github.com/Computational-Rheology/OpenFOAMCourse_PPS41/raw/refs/heads/main/caseFiles.zip`
5. `>> unzip CaseFiles.zip`
6. `>> cd case 11`
7. `>> code .`
8. Check the case structure with the instructor, in folders *0*, *constant* and *system*



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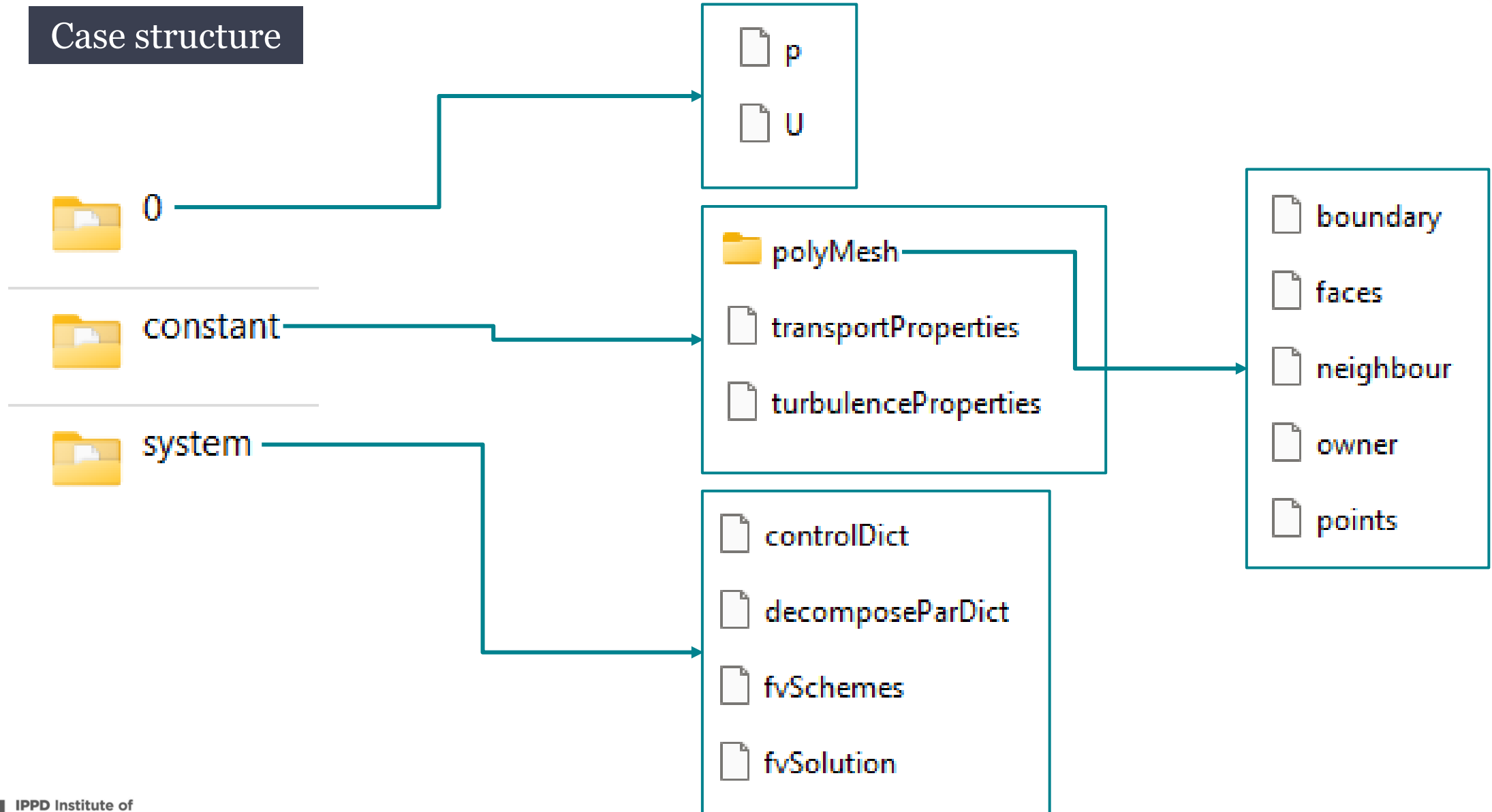


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OpenFOAM 1st case study

Case structure



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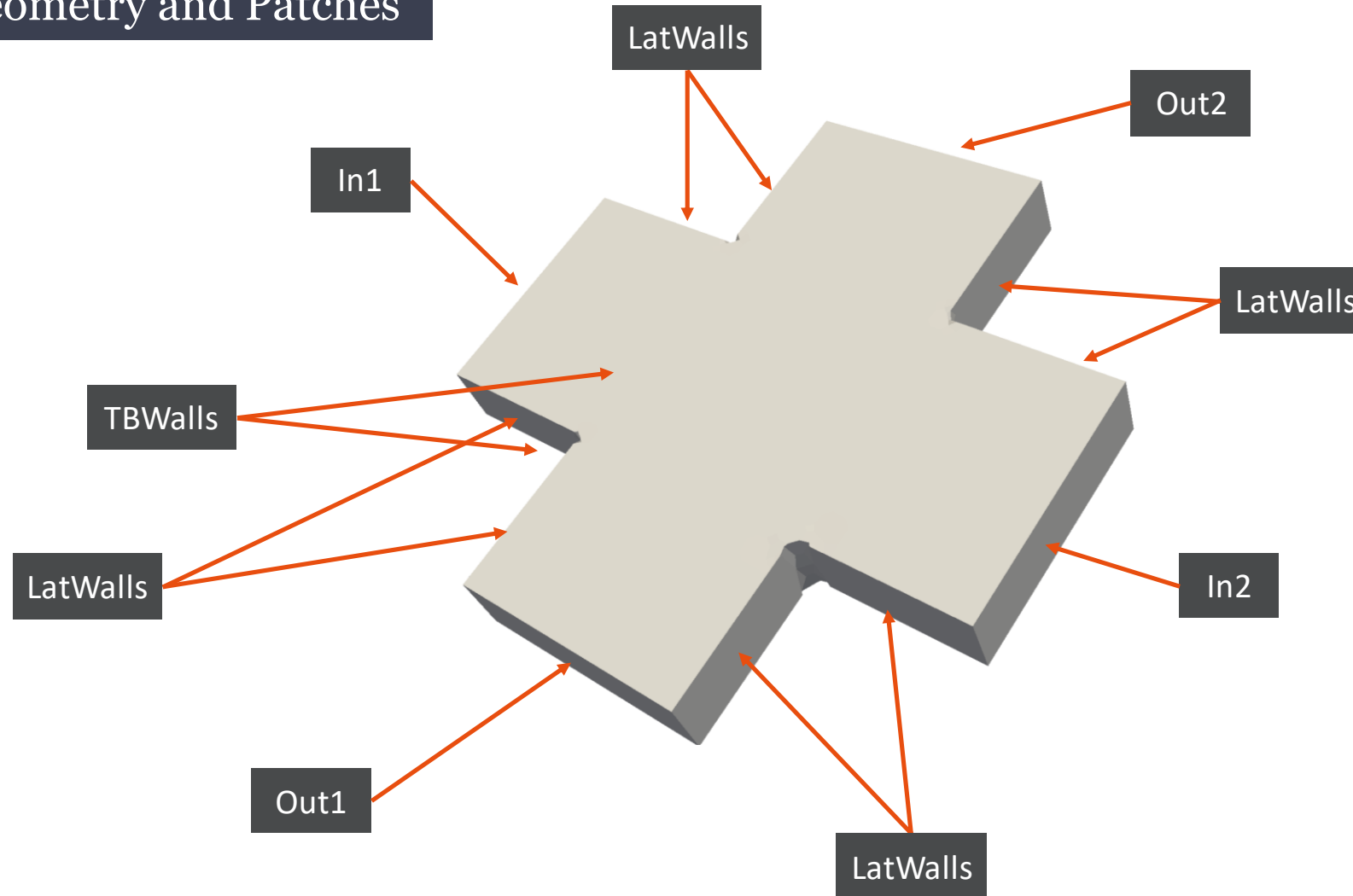
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OpenFOAM 1st case study

Geometry and Patches



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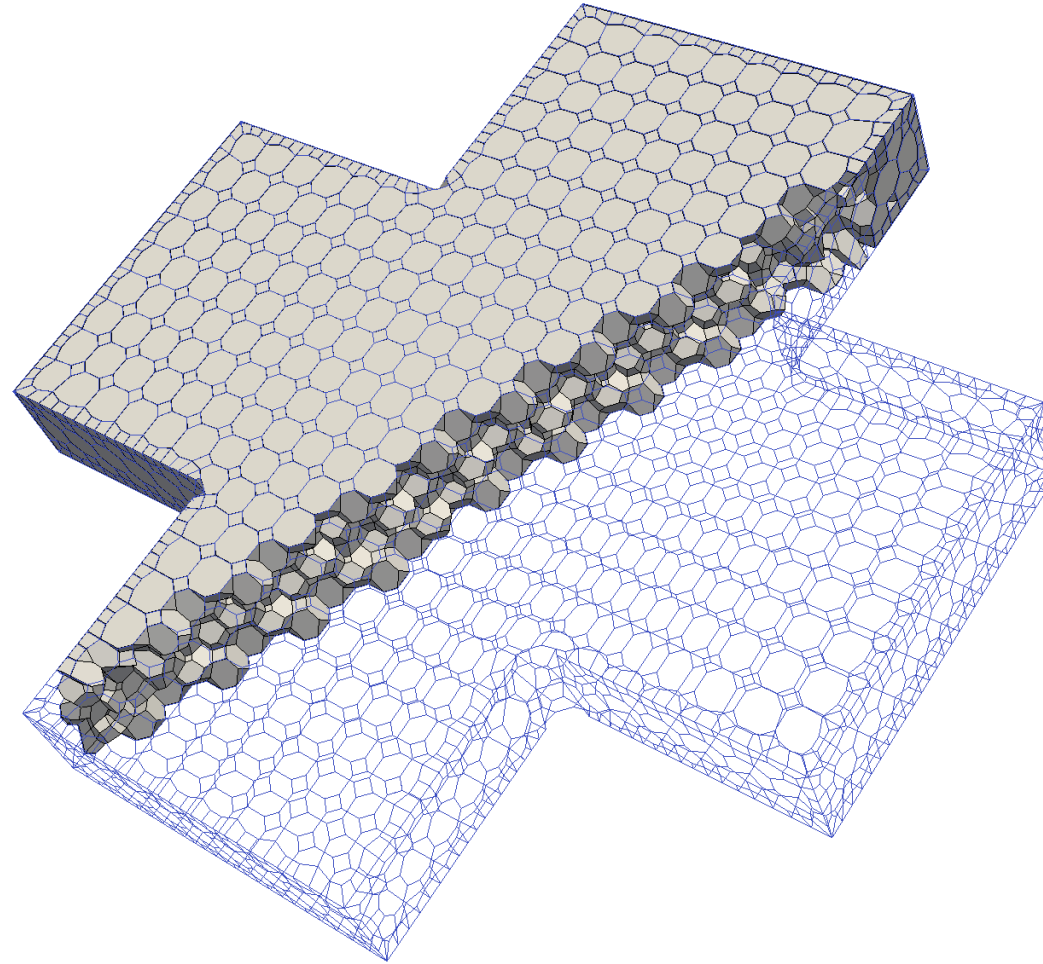
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OpenFOAM 1st case study

Mesh



Session P2



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OpenFOAM 1st case study

File o/U

.....
dimensions [0 1 -1 0 0 0 0];

internalField uniform (0 0 0);

boundaryField
{

In1

{
 type fixedValue;
 value uniform (0.01 0 0);
}

In2

{
 type fixedValue;
 value uniform (-0.01 0 0);
}

Out1

{
 type zeroGradient;
}

.....

No.	Property
1	Mass
2	Length
3	Time
4	Temperature
5	Quantity
6	Current
7	Luminous intensity



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OpenFOAM 1st case study

File o/p

.....

```
boundaryField
{
```

```
    In1
```

```
    {
        type        zeroGradient;
    }
```

```
    In2
```

```
    {
        type        zeroGradient;
    }
```

```
    Out1
```

```
    {
        type        fixedValue;
        value        uniform 0;
    }
```

```
.....
```



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File constant/transportProperties

.....

transportModel Newtonian;

nu nu [0 2 -1 0 0 0 0] 1e-05;

.....



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OpenFOAM 1st case study

File system/controlDict

```
.....  
application simpleFoam;  
  
startFrom      startTime;  
  
startTime      0;  
  
stopAt         endTime;  
  
endTime        1000;  
  
deltaT         1;  
  
writeControl    timeStep;  
  
writeInterval   500;  
  
purgeWrite      0;  
  
.....
```



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File system/fvSchemes

ddtSchemes

```
{  
  default      steadyState;  
}
```

gradSchemes

```
{  
  default      Gauss linear;  
}
```

divSchemes

```
{  
  default      none;  
  div(phi,U)   bounded Gauss linearUpwind grad(U);  
  ...  
  div((nuEff*dev2(T(grad(U)))) Gauss linear;  
  div(nonlinearStress) Gauss linear;  
}
```

laplacianSchemes

```
{  
  default      Gauss linear corrected;  
}  
....
```



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OpenFOAM 1st case study

File system/fvSolution

```
....  
solvers  
{  
  p  
  {  
    solver      GAMG;  
    tolerance   1e-07;  
    relTol      0.1;  
    smoother    GaussSeidel;  
  }  
....  
SIMPLE  
{  
  nNonOrthogonalCorrectors 2;  
  consistent    yes;  
  
  residualControl  
  {  
    p          1e-6;  
    U          1e-6;  
    "(k|epsilon|omega|f|v2)" 1e-6;  
  }  
}  
.....
```



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OpenFOAM 1st case study

1. >> simpleFoam #run the solver and output screen
2. >> rm -rf 112 #remove the lastTime results folder
3. >> simpleFoam >log.simpleFoam #run the solver and output to log.simpleFoam file
4. >> rm -rf 112 #remove the lastTime results folder
5. >> simpleFoam >log.simpleFoam & #run the solver in background and output to log.simpleFoam file
6. Check the contents of the logfile

OpenFOAM 1st case study

1. >> touch x.foam
2. >> explorer.exe .
3. Open *x.foam* file with paraview
4. Visualize the results
 - a) Show/hide mesh
 - b) FV results or interpolated
 - c) Rescale color map
 - d) Show/hide interior cells and patches
 - e) Stream tracer (source line and point cloud)
 - f) Contour, clip, slice, threshold and glyph
5. Adapt the velocity boundary conditions
6. Visualize the results in paraview



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OpenFOAM 1st case study

1. Check the contents of the file system/decomposeParDict with the instructor
2. The banana trick
3. >> decomposePar
4. Use the scotch method to decompose
5. >> mpirun -np 4 simpleFoam -parallel > log.simpleFoamP
6. >> simpleFoam > log.simpleFoam
7. Compare the execution time of both runs



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